

Acknowledgments We thank Deepak Pathak, Piotr Dollár, Ranjay Krishna, Animesh Garg, and Danfei Xu for helpful comments and discussion.

References

- [1] A. Agrawal, D. Batra, and D. Parikh. Analyzing the behavior of visual question answering models. In *EMNLP*, 2016. 1
- [2] J. Andreas, M. Rohrbach, T. Darrell, and D. Klein. Learning to compose neural networks for question answering. In *NAACL*, 2016. 1, 2, 4, 5, 8, 11
- [3] J. Andreas, M. Rohrbach, T. Darrell, and D. Klein. Neural module networks. In *CVPR*, 2016. 1, 2, 4, 8, 11
- [4] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. Zitnick, and D. Parikh. VQA: Visual question answering. In *ICCV*, 2015. 1, 2, 4, 5, 11
- [5] Y. Bengio, A. Courville, and P. Vincent. Representation learning: A review and new perspectives. *TPAMI*, 35(8):1798–1828, 2014. 8
- [6] Blender Online Community. *Blender - a 3D modelling and rendering package*. Blender Foundation, Blender Institute, Amsterdam, 2016. 3
- [7] J. Chen, P. Kuznetsova, D. Warren, and Y. Choi. Deja image-captions: A corpus of expressive image descriptions in repetition. In *NAACL*, 2015. 2
- [8] A. Farhadi, M. Hejrati, A. Sadeghi, P. Young, C. Rashtchian, J. Hockenmaier, and D. Forsyth. Every picture tells a story: Generating sentences for images. In *ECCV*, 2010. 2
- [9] A. Fukui, D. H. Park, D. Yang, A. Rohrbach, T. Darrell, and M. Rohrbach. Multimodal compact bilinear pooling for visual question answering and visual grounding. In *arXiv:1606.01847*, 2016. 1, 4, 5
- [10] H. Gao, J. Mao, J. Zhou, Z. Huang, L. Wang, and W. Xu. Are you talking to a machine? Dataset and methods for multilingual image question answering. In *NIPS*, 2015. 1, 2, 4
- [11] Y. Gao, O. Beijbom, N. Zhang, and T. Darrell. Compact bilinear pooling. In *CVPR*, 2016. 5
- [12] D. Geman, S. Geman, N. Hallonquist, and L. Younes. Visual Turing test for computer vision systems. *Proceedings of the National Academy of Sciences*, 112(12):3618–3623, 2015. 2
- [13] A. Graves, G. Wayne, M. Reynolds, T. Harley, I. Danihelka, A. Grabska-Barwinska, S. Colmenarejo, E. Grefenstette, T. Ramalho, J. Agapiou, A. Badia, K. Hermann, Y. Zwols, G. Ostrovski, A. Cain, H. King, C. Summerfield, P. Blunsom, K. . Kavukcuoglu, and D. Hassabis. Hybrid computing using a neural network with dynamic external memory. *Nature*, 2016. 8
- [14] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 5
- [15] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. 2, 5
- [16] A. Jabri, A. Joulin, and L. van der Maaten. Revisiting visual question answering baselines. In *ECCV*, 2016. 1, 4
- [17] J. Johnson, R. Krishna, M. Stark, L.-J. Li, D. A. Shamma, M. S. Bernstein, and L. Fei-Fei. Image retrieval using scene graphs. In *CVPR*, 2015. 3
- [18] A. Joulin and T. Mikolov. Inferring algorithmic patterns with stack-augmented recurrent nets. In *NIPS*, 2015. 8
- [19] S. Kazemzadeh, V. Ordonez, M. Matten, and T. Berg. Referitgame: Referring to objects in photographs of natural scenes. In *EMNLP*, 2014. 2
- [20] D. Kingma and J. Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015. 5
- [21] R. Krishna, Y. Zhu, O. Groth, J. Johnson, K. Hata, J. Kravitz, S. Chen, Y. Kalantidis, L. Jia-Li, D. Shamma, M. Bernstein, and L. Fei-Fei. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *IJCV*, 2016. 1, 2, 3
- [22] H. J. Levesque, E. Davis, and L. Morgenstern. The Winograd schema challenge. In *AAAI Spring Symposium: Logical Formalizations of Commonsense Reasoning*, volume 46, page 47, 2011. 2
- [23] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollar, and C. Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014. 2
- [24] J. Lu, J. Yang, D. Batra, and D. Parikh. Hierarchical question-image co-attention for visual question answering. In *NIPS*, 2016. 1, 2, 4
- [25] L. Ma, Z. Lu, and H. Li. Learning to answer questions from image using convolutional neural network. In *AAAI*, 2016. 4
- [26] M. Malinowski and M. Fritz. A multi-world approach to question answering about real-world scenes based on uncertain input. In *NIPS*, 2014. 1, 2
- [27] M. Malinowski and M. Fritz. Towards a visual Turing challenge. In *NIPS 2014 Workshop on Learning Semantics*, 2014. 2
- [28] M. Malinowski, M. Rohrbach, and M. Fritz. Ask your neurons: A neural-based approach to answering questions about images. In *ICCV*, 2015. 2, 4
- [29] T. Mikolov, K. Chen, G. Corrado, and J. Dean. Efficient estimation of word representations in vector space. In *arXiv 1301.3781*, 2013. 5
- [30] O. Pfungst. *Clever Hans (The horse of Mr. von Osten): A contribution to experimental animal and human psychology*. Henry Holt, New York, 1911. 1
- [31] A. Ray, G. Christie, M. Bansal, D. Batra, and D. Parikh. Question relevance in vqa: Identifying non-visual and false-premise questions. In *EMNLP*, 2016. 1
- [32] M. Ren, R. Kiros, and R. Zemel. Exploring models and data for image question answering. In *NIPS*, 2015. 1, 2, 4
- [33] K. Shih, S. Singh, and D. Hoiem. Where to look: Focus regions for visual question answering. In *CVPR*, 2016. 4
- [34] N. Srivastava, G. E. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *JMLR*, 15(1):1929–1958, 2014. 5
- [35] B. Sturm. A simple method to determine if a music information retrieval system is a horse. *IEEE Transactions on Multimedia*, 16(6):1636–1644, 2014. 1
- [36] B. Sturm. Horse taxonomy and taxidermy. *HORSE2016*, 2016. 1
- [37] M. Tapaswi, Y. Zhu, R. Stiefelhagen, A. Torralba, R. Urtasun, and S. Fidler. Movieqa: Understanding stories in movies through question-answering. In *CVPR*, 2016. 2

- [38] J. Weston, A. Bordes, S. Chopra, A. Rush, B. van Merriënboer, A. Joulin, and T. Mikolov. Towards ai-complete question answering: A set of prerequisite toy tasks. In *ICLR*, 2016. 2
- [39] J. Weston, S. Chopra, and A. Bordes. Memory networks. In *ICLR*, 2015. 8
- [40] T. Winograd. *Understanding Natural Language*. Academic Press, 1972. 2
- [41] Q. Wu, C. Shen, A. van den Hengel, P. Wang, and A. Dick. Image captioning and visual question answering based on attributes and their related external knowledge. In *arXiv 1603.02814*, 2016. 4
- [42] C. Xiong, S. Merity, and R. Socher. Dynamic memory networks for visual and textual question answering. *ICML*, 2016. 4, 8
- [43] H. Xu and K. Saenko. Ask, attend, and answer: Exploring question-guided spatial attention for visual question answering. In *ECCV*, 2016. 4
- [44] Z. Yang, X. He, J. Gao, L. Deng, and A. Smola. Stacked attention networks for image question answering. In *CVPR*, 2016. 1, 2, 4, 5, 8
- [45] P. Young, A. Lai, M. Hodosh, and J. Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. In *TACL*, pages 67–78, 2014. 2
- [46] L. Yu, E. Park, A. Berg, and T. Berg. Visual madlibs: Fill in the blank image generation and question answering. In *ICCV*, 2015. 1, 2
- [47] P. Zhang, Y. Goyal, D. Summers-Stay, D. Batra, and D. Parikh. Yin and yang: Balancing and answering binary visual questions. In *CVPR*, 2016. 1, 2
- [48] B. Zhou, Y. Tian, S. Sukhbataar, A. Szlam, and R. Fergus. Simple baseline for visual question answering. In *arXiv:1512.02167*, 2015. 1, 4, 5
- [49] Y. Zhu, O. Groth, M. Bernstein, and L. Fei-Fei. Visual7w: Grounded question answering in images. In *CVPR*, 2016. 1, 2, 4
- [50] C. Zitnick and D. Parikh. Bringing semantics into focus using visual abstraction. In *CVPR*, 2013. 2